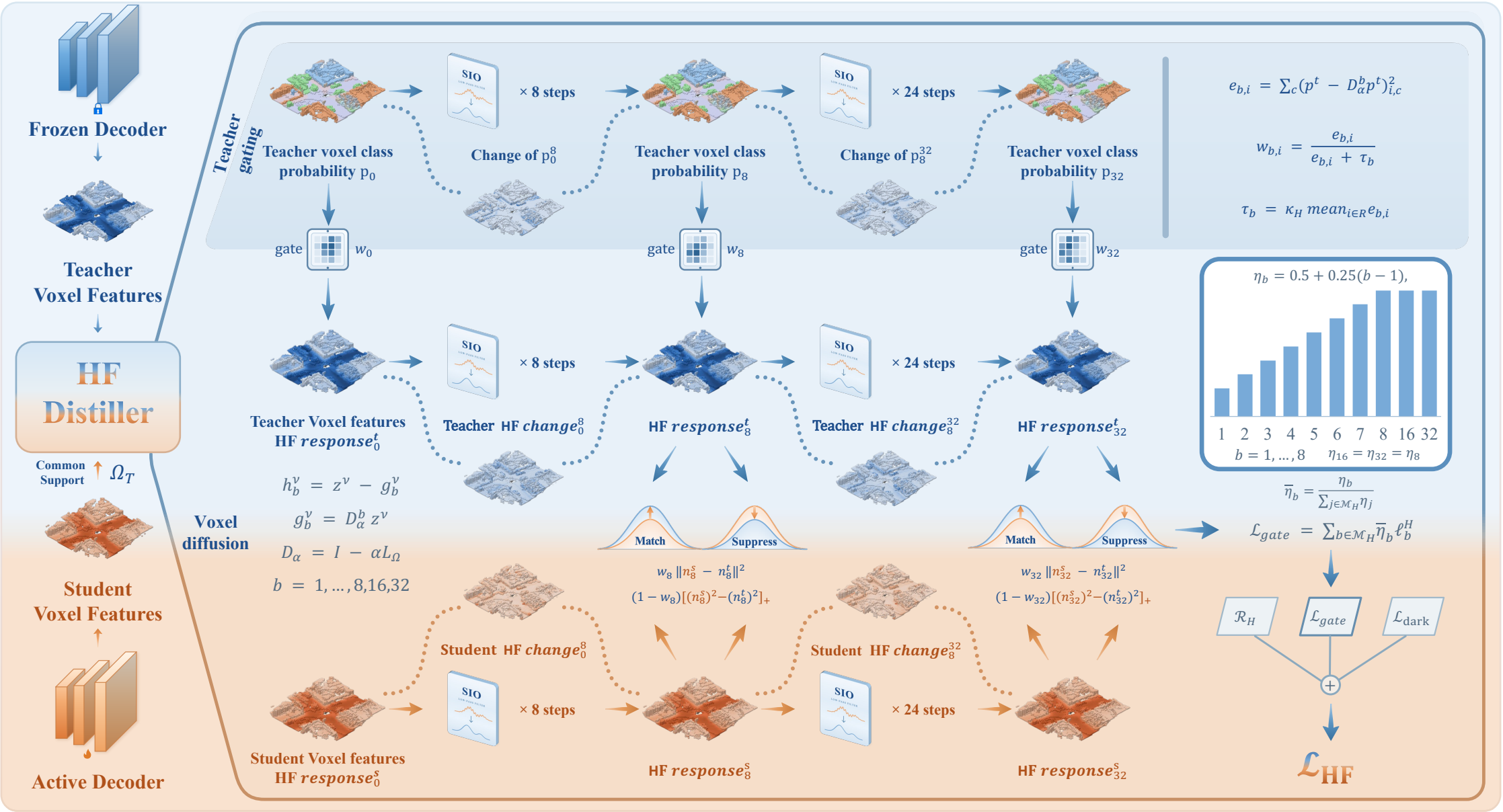

Frozen Decoder
Teacher Voxel Features
HF Distiller
Common Support Ω_T
Student Voxel Features
Active Decoder
Teacher gating
Teacher voxel class probability p_0
SIO
× 8 steps
Change of p_0^8
Teacher voxel class probability p_8
Change of p_8^{32}
× 24 steps
Teacher voxel class probability p_{32}
gate w_0
gate w_8
gate w_{32}
Teacher Voxel features HF response$_0^t$
Teacher HF change$_0^8$
HF response$_8^t$
Teacher HF change$_8^{32}$
HF response$_{32}^t$
Voxel diffusion
$h_b^v = z^v - g_b^v$
$g_b^v = D_\alpha^b z^v$
$D_\alpha = I - \alpha L_\Omega$
$b = 1, \dots, 8, 16, 32$
Match
Suppress
$w_8 \|n_8^s - n_8^t\|^2$
$(1-w_8)[(n_8^s)^2-(n_8^t)^2]_+$
$w_{32} \|n_{32}^s - n_{32}^t\|^2$
$(1-w_{32})[(n_{32}^s)^2-(n_{32}^t)^2]_+$
Student HF change$_0^8$
Student HF change$_8^{32}$
Student Voxel features HF response$_0^s$
HF response$_8^s$
HF response$_{32}^s$
$e_{b,i} = \sum_c (p^t - D_\alpha^b p^t)_{i,c}^2$
$w_{b,i} = \frac{e_{b,i}}{e_{b,i} + \tau_b}$
$\tau_b = \kappa_H \, mean_{i \in R} e_{b,i}$
$\eta_b = 0.5 + 0.25(b-1)$,
1 2 3 4 5 6 7 8 16 32
$b = 1, \dots, 8$
$\eta_{16} = \eta_{32} = \eta_8$
$\overline{\eta}_b = \frac{\eta_b}{\sum_{j \in \mathcal{M}_H} \eta_j}$
$\mathcal{L}_{gate} = \sum_{b \in \mathcal{M}_H} \overline{\eta}_b \ell_b^H$
$\mathcal{R}_H$
$\mathcal{L}_{gate}$
$\mathcal{L}_{dark}$
$\mathcal{L}_{HF}$